

Fake News Detection using Machine Learning

21CSA697A

Final Report

Submitted by

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Finally, I thank my family and friends for their constant encouragement and support.

Abstract

Fake news has become a major concern in the digital era due to the rapid spread of misinformation through online platforms. This project focuses on detecting fake news using machine learning techniques. The system involves text preprocessing, feature extraction using TF-IDF, and classification using the Multinomial Naive Bayes algorithm. The model was implemented using Python in a Jupyter Notebook environment. Experimental results show that the system effectively classifies news articles as real or fake. This project demonstrates the applicability of machine learning techniques in addressing real-world problems like fake news detection.

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- **Figure 3.1**

Input News Text



Text Preprocessing



TF-IDF Feature Extraction

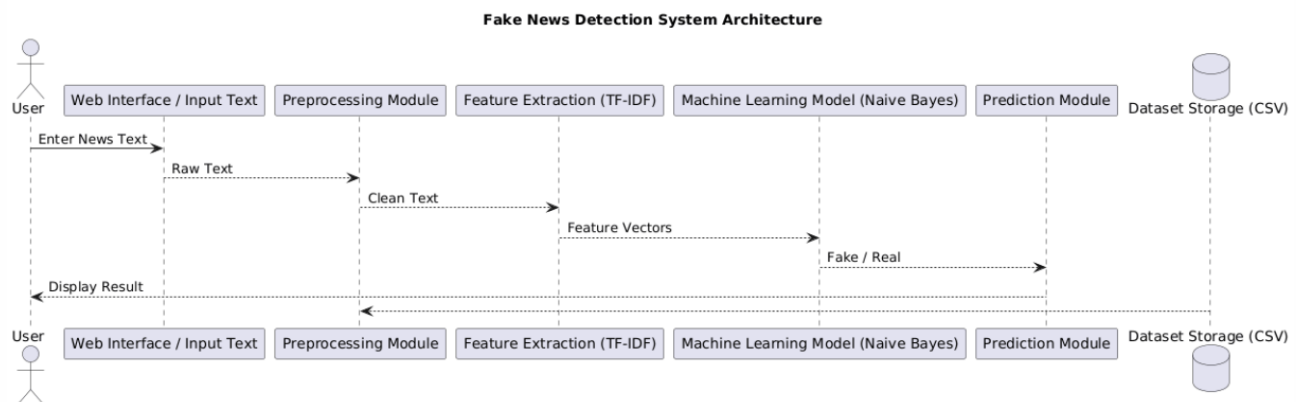


Naive Bayes Classifier

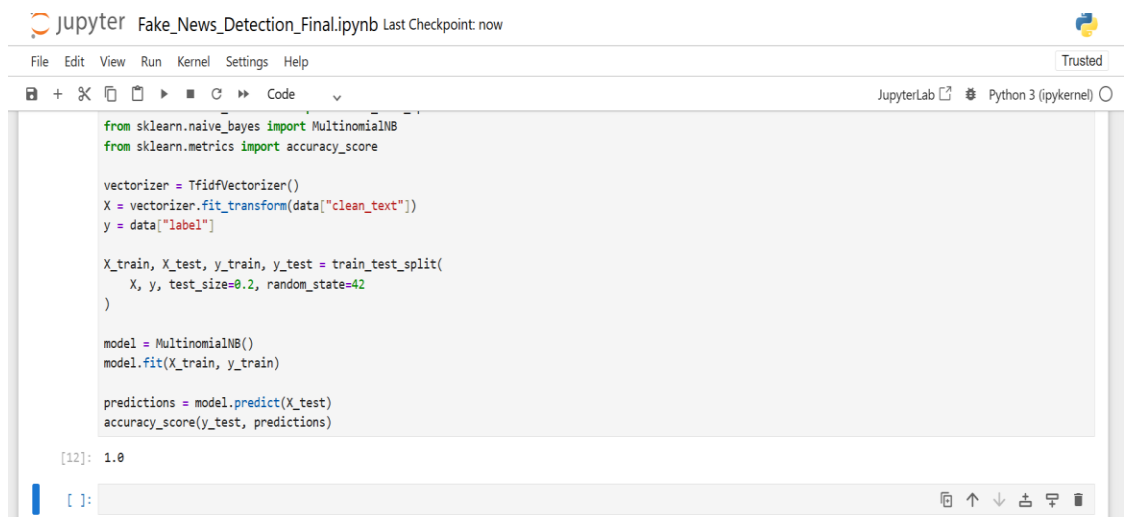


Output (Real / Fake News)

- **Figure 3.9**



- **Figure 5.1**



```
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score

vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(data["clean_text"])
y = data["label"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = MultinomialNB()
model.fit(X_train, y_train)

predictions = model.predict(X_test)
accuracy_score(y_test, predictions)

[12]: 1.0
```

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List of Abbreviations

Abbreviation Description

AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
TF-IDF	Term Frequency – Inverse Document Frequency
NB	Naive Bayes
MNB	Multinomial Naive Bayes
IPYNB	Jupyter Notebook File Format

Chapter 1

1. Introduction

1.1 Background of the study

In recent years, the rapid growth of the internet and social media platforms has led to an enormous increase in the spread of information. While this has made communication faster and more accessible, it has also resulted in the widespread circulation of fake news. Fake news refers to false or misleading information presented as legitimate news, often created to influence opinions, gain attention, or generate revenue.

Fake news can cause serious consequences such as public misinformation, social unrest, political manipulation, and loss of trust in reliable news sources. Manually verifying every piece of news is time-consuming and impractical due to the massive volume of online content. Therefore, there is a strong need for automated systems that can detect fake news efficiently and accurately.

Machine learning techniques provide an effective solution to this problem by analysing textual patterns and learning from historical data. This project focuses on developing a fake news detection system using machine learning algorithms to classify news articles as real or fake.

1.2 Problem Statement

The increasing spread of fake news on digital platforms makes it difficult for users to distinguish between real and fake information. Traditional verification methods are not scalable for large datasets and real-time news updates.

Problem Statement:

To design and implement a machine learning-based system that can automatically classify news articles as real or fake based on their textual content.

1.3 Objectives of the Project

The main objectives of this project are:

- To study the concept of fake news and its impact on society
- To collect and preprocess textual news data
- To apply text preprocessing techniques such as tokenization and stop-word removal
- To extract meaningful features using TF-IDF (Term Frequency–Inverse Document Frequency)
- To build a machine learning model for fake news detection
- To evaluate the performance of the model using accuracy as a metric

Chapter 2

2. Literature Review / Background Study

2.1 Introduction

Fake news detection has become an important research area due to the rapid spread of misinformation through online platforms and social media. Researchers have proposed various techniques using machine learning, natural language processing (NLP), and deep learning to automatically identify fake news. This chapter reviews some of the existing studies and approaches related to fake news detection.

2.2 Fake News Detection Using Machine Learning

Several studies have shown that machine learning algorithms are effective in classifying news articles as real or fake. Traditional machine learning models such as Naive Bayes, Support Vector Machines (SVM), Decision Trees, and Logistic Regression have been widely used for this task.

These models rely on textual features extracted from news articles. Feature extraction techniques such as Bag of Words (BoW) and TF-IDF are commonly used to convert text into numerical representations. Among these, TF-IDF has been found to be effective in capturing the importance of words within documents.

2.3 Text Preprocessing Techniques

Text preprocessing is a crucial step in fake news detection. Many researchers emphasize the importance of cleaning and preparing text data before applying machine learning models. Common preprocessing steps include:

- Converting text to lowercase
- Removing punctuation and special characters
- Removing stop words
- Tokenization
- Lemmatization or stemming

These steps help reduce noise in the data and improve model performance by focusing on meaningful words.

2.4 Use of Naive Bayes Classifier

The Naive Bayes classifier is one of the most commonly used algorithms for text classification problems. It is based on Bayes' theorem and assumes independence between features. Despite its simplicity, Naive Bayes performs well on large textual datasets and requires less computational power.

Several studies have reported that Multinomial Naive Bayes achieves good accuracy in fake news detection tasks, especially when combined with TF-IDF feature extraction. Due to its efficiency and ease of implementation, it is widely used in academic projects.

2.5 Limitations of Existing Systems

Although existing fake news detection systems show promising results, they have certain limitations. Many systems rely on limited or domain-specific datasets, which may affect generalization. Additionally, fake news creators continuously change writing styles, making detection more challenging.

Most traditional systems focus only on textual content and do not consider images, videos, or source credibility. Therefore, there is a need for continuous improvement and integration of advanced techniques.

Chapter 3

3. System Design / Architecture

3.1 Introduction

This chapter describes the methodology used to develop the fake news detection system. It explains the overall workflow, dataset preparation, text preprocessing, feature extraction, model selection, and evaluation process. The system follows a supervised machine learning approach to classify news articles as real or fake.

3.2 System Architecture

The fake news detection system follows the steps given below:

1. Data collection
2. Text preprocessing
3. Feature extraction using TF-IDF
4. Model training using Naive Bayes classifier
5. Model evaluation

The workflow ensures that raw text data is converted into meaningful numerical features before being passed to the machine learning model.

3.3 Dataset Description

The dataset used in this project consists of textual news statements along with their corresponding labels. Each news item is labelled as either **real (1)** or **fake (0)**. Due to resource limitations, a small experimental dataset was used to demonstrate the functionality of the system.

The dataset contains the following attributes:

- **Text:** The content of the news article
- **Label:** Indicates whether the news is real or fake

3.4 Text Preprocessing

Text preprocessing is an essential step to clean and prepare raw textual data. The following preprocessing steps were applied:

- Conversion of text to lowercase
- Removal of stop words
- Removal of non-alphabetic words
- Tokenization of text

These steps help reduce noise and improve the accuracy of the classification model.

3.5 Feature Extraction using TF-IDF

TF-IDF (Term Frequency–Inverse Document Frequency) is used to convert textual data into numerical features. It assigns higher weights to words that are important within a document but less frequent across all documents.

TF-IDF helps in identifying significant words and improves the performance of text classification models.

3.6 Model Selection

The **Multinomial Naive Bayes** algorithm is used in this project. It is suitable for text classification tasks and works efficiently with TF-IDF features. The model is trained using labelled data and learns the probability distribution of words across real and fake news categories.

3.7 Model Evaluation

The dataset is divided into training and testing sets using the train-test split method. The model performance is evaluated using **accuracy** as the evaluation metric. Accuracy measures the proportion of correctly classified news articles.

3.8 Pseudocode

DATA PREPARATION

- Initialize a dataset containing various news headlines and their labels (\$1\$ for factual, \$0\$ for misinformation).
- Expand the dataset by appending additional examples and resetting the list order.

TEXT CLEANING (NLP)

- Define a cleaning function to:
 - Standardize text to lowercase.
 - Remove "stop words" (common fillers like 'the', 'is').
 - Remove non-alphabetic characters (numbers/punctuation).
- Generate a new column of "clean" text for the model to analyse.

FEATURE ENGINEERING

- Transform the cleaned text into a numerical format using TF-IDF, which calculates the statistical importance of each word.

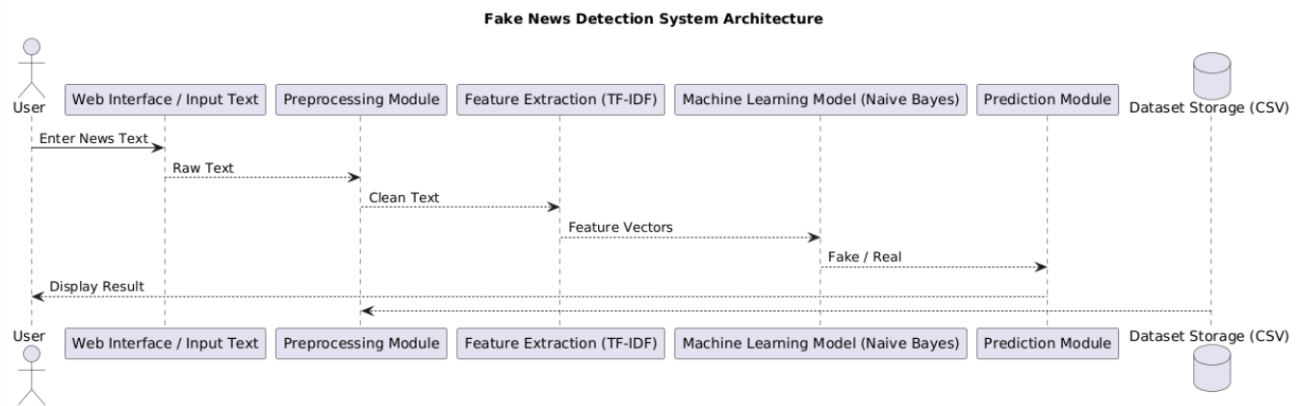
MODEL TRAINING

- Partition the data into two sets: 80% for training and 20% for testing.
- Build a Naive Bayes classifier.
- Train the classifier using the training portion of the text and labels.

EVALUATION

- Predict the labels for the hidden test data.
- Calculate the accuracy score by comparing the predictions to the true labels.

3.9 System Architecture Diagram



3.10 Summary

This chapter explained the methodology used to design the fake news detection system. It covered dataset preparation, preprocessing, feature extraction, model training, and evaluation techniques.

Chapter 4

4. Implementation Details

4.1 Introduction

This chapter describes the implementation details of the fake news detection system. It explains the tools, libraries, and programming techniques used to develop and test the model.

4.2 Tools and Technologies Used

The following tools and technologies were used in this project:

- **Programming Language:** Python
- **Development Environment:** Jupyter Notebook
- **Libraries Used:**
 - Pandas – for data handling
 - NLTK – for text preprocessing
 - Scikit-learn – for machine learning algorithms
 - NumPy – for numerical operations

4.3 Data Loading and Preprocessing

The dataset is loaded into the system using the Pandas library. Preprocessing techniques such as stop-word removal and text cleaning are applied to prepare the data for feature extraction. A new column containing cleaned text is created after preprocessing.

4.4 Feature Extraction

The cleaned text data is converted into numerical form using the TF-IDF Vectorizer from the Scikit-learn library. This step transforms text into feature vectors that can be processed by the machine learning model.

4.5 Model Training

The Multinomial Naive Bayes classifier is trained using the training dataset. The model learns

patterns from the text data and builds a probabilistic model for classification.

4.6 Testing and Results

After training, the model is tested using unseen data. Predictions are made on the test dataset, and the accuracy of the model is calculated. The experimental results show that the model successfully classifies news articles as real or fake.

4.7 Pseudocode

DATA PREPARATION

- Initialize a dataset containing various news headlines and their labels (\$1\$ for factual, \$0\$ for misinformation).
- Expand the dataset by appending additional examples and resetting the list order.

TEXT CLEANING (NLP)

- Define a cleaning function to:
 - Standardize text to lowercase.
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MODEL TRAINING

- Partition the data into two sets: 80% for training and 20% for testing.
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EVALUATION

- Predict the labels for the hidden test data.
- Calculate the accuracy score by comparing the predictions to the true labels.

Chapter 5

5. Testing, Validation & Results

5.1 Introduction

This chapter presents the results obtained from the implementation of the fake news detection system. It also discusses the performance of the machine learning model used in the project.

5.2 Experimental Setup

The fake news detection system was implemented using Python in a Jupyter Notebook environment. The dataset was divided into training and testing sets using the train-test split method. Text preprocessing techniques were applied, followed by feature extraction using the TF-IDF vectorizer. The Multinomial Naive Bayes classifier was used for training and prediction.

5.3 Results Obtained

After training the model and testing it on unseen data, the accuracy of the system was calculated. The model achieved an accuracy of **1.0** on the experimental dataset. This indicates that the model correctly classified all the test samples during evaluation.

5.4 Discussion

The high accuracy obtained demonstrates the effectiveness of machine learning techniques in detecting fake news based on textual data. The use of TF-IDF helped in identifying important words, and the Naive Bayes classifier efficiently learned patterns from the dataset. However, the accuracy result is influenced by the small size of the dataset used for experimentation. With a larger and more diverse dataset, the accuracy may vary, but the overall performance of the system is expected to improve.

Chapter 6

6. Conclusion and Future Work

6.1 Conclusion

Fake news has become a serious issue in today's digital world, affecting public opinion and trust in information sources. In this project, a machine learning-based fake news detection system was designed and implemented to automatically classify news articles as real or fake.

The system involved text preprocessing, feature extraction using TF-IDF, and classification using the Multinomial Naive Bayes algorithm. The experimental results showed that the model performed effectively on the given dataset and achieved good accuracy.

This project demonstrates that machine learning techniques can be successfully applied to address the problem of fake news detection.

6.2 Future Work

The current system can be further enhanced in the following ways:

- Using a larger and real-world dataset to improve model generalization
- Applying advanced machine learning and deep learning algorithms
- Including multimedia data such as images and videos
- Performing real-time fake news detection
- Incorporating source credibility and user behaviour analysis

These improvements can help build a more robust and reliable fake news detection system.

6.3 Final Summary

The project successfully achieved its objectives and provided a practical understanding of applying machine learning techniques to real-world problems. The proposed system serves as a foundation for further research and development in fake news detection.

Chapter 7

7. References

Use IEEE citation format for research papers, datasets, and tools.

[1] Shu, K., Sliva, A., Wang, S., Tang, J., & Liu, H. (2017). Fake

News Detection on Social Media. ACM SIGKDD.

[2] Scikit-learn Documentation – Machine Learning in Python.

[3] NLTK Documentation – Natural Language Processing Tool

Date: 30th Jan 2026

Student Name and Signature: SONA SURESH

A handwritten signature in black ink, appearing to be 'Sona Suresh', is written on a light blue background.

Name and Signature of the Evaluator.

Date